Ritz Method — Gelfand - Fomin Problem 8.2

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1 Problem Statement

2. Use the Ritz method to find an approximate solution of the problem of minimizing the functional

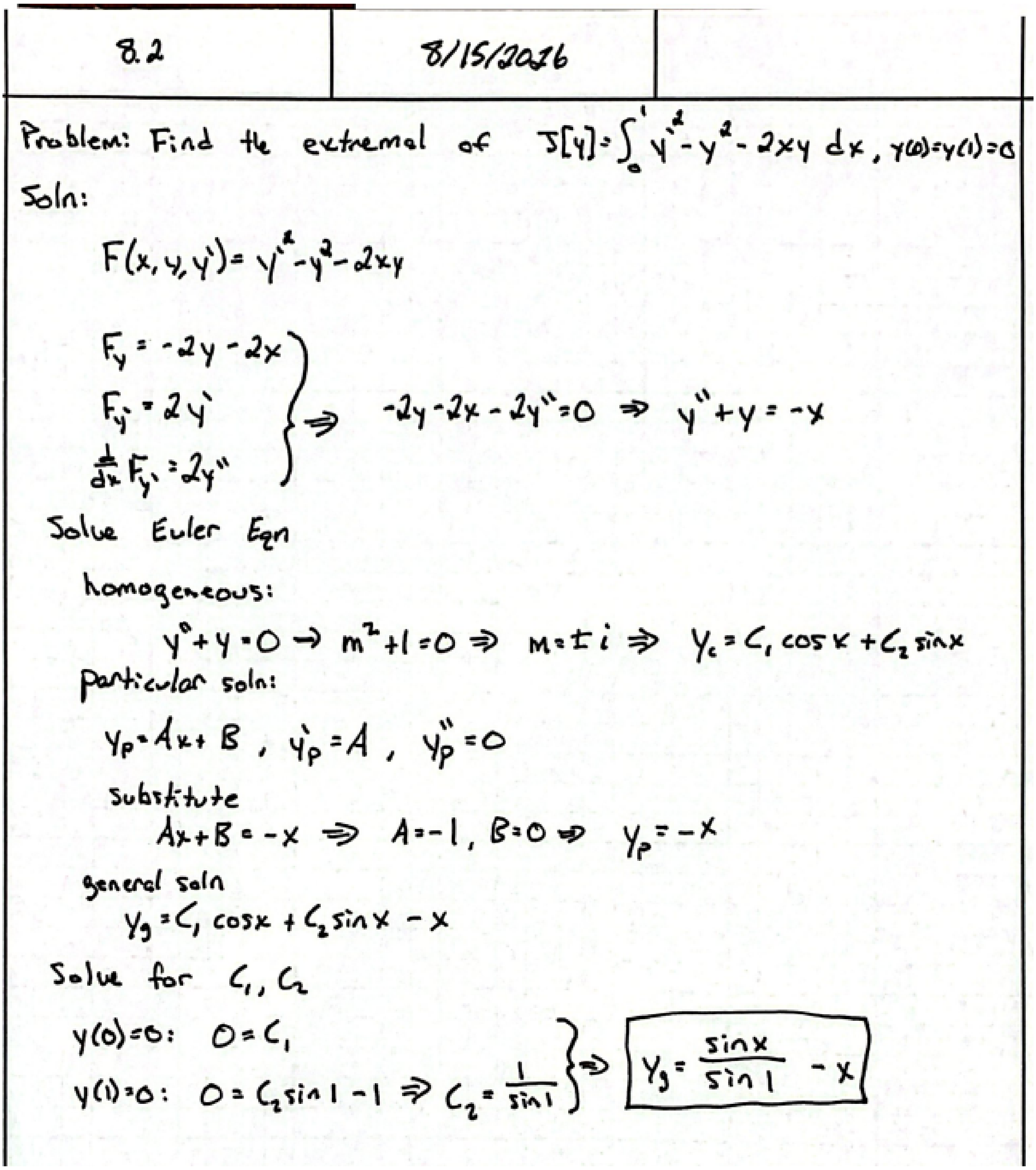
$$J[y] = \int_0^1 (y'^2 - y^2 - 2xy)\,dx, \qquad y(0) = y(1) = 0,$$

and compare the answer with the exact solution.

Figure 1: Problem 8.2

2 Solution

2.1 Exact Solution



8.2 | 8/15/2016

Problem: Find the extremal of $J[y] = \int_0^1 y'^2 - y^2 - 2xy \, dx$, $y(0) = y(1) = 0$

Soln:

$$F(x, y, y') = y'^2 - y^2 - 2xy$$

$$\left.\begin{array}{l} F_y = -2y - 2x \\ F_{y'} = 2y' \\ \frac{d}{dx} F_{y'} = 2y'' \end{array}\right\} \Rightarrow \quad -2y - 2x - 2y'' = 0 \Rightarrow y'' + y = -x$$

Solve Euler Eqn

homogeneous:

$$y'' + y = 0 \rightarrow m^2 + 1 = 0 \Rightarrow m = \pm i \Rightarrow y_c = C_1 \cos x + C_2 \sin x$$

particular soln:

$$y_p = Ax + B, \quad y_p' = A, \quad y_p'' = 0$$

substitute

$$Ax + B = -x \Rightarrow A = -1, \; B = 0 \Rightarrow y_p = -x$$

general soln

$$y_g = C_1 \cos x + C_2 \sin x - x$$

Solve for C_1, C_2

$$\left.\begin{array}{l} y(0) = 0: \quad 0 = C_1 \\ y(1) = 0: \quad 0 = C_2 \sin 1 - 1 \Rightarrow C_2 = \frac{1}{\sin 1} \end{array}\right\} \Rightarrow \boxed{y_g = \frac{\sin x}{\sin 1} - x}$$

Figure 2: Problem 8.2

We will define the julia function of this for comparison against the approximations

```
y_exact(x) = sin(x) / sin(1) - x
have_exact = !isnothing(y_exact)
```

2.2 Ritz Approximations

We will use the `RitzMethod` package

```
using RitzMethod
import Optim
import Plots

# Define problem and basis
integrand(x, y, y_prime) = y_prime^2 - y^2 - 2 * x * y
domain = Interval(0.0, 1.0)
full_basis = [x -> x^k * (1 - x) for k in 1:3]

function solve_ritz(n_terms)
    minimizefunctional(
        integrand, domain, full_basis[1:n_terms];
        num_quad_nodes=5,
        method=Optim.BFGS(),
        options=Optim.Options(iterations=50, g_abstol=1e-20, x_reltol=1e-8),
    )
end

results = [solve_ritz(n) for n in 1:3]
```

2.2.1 One-Term Approximation

```
println(summarize(results[1]))
```

```
RitzResult:
  Domain:          [0.0, 1.0]
  Coefficients:    [0.2777777777777778]
  Minimized value: -0.02314814814814817
  Converged:       true
```

2.2.2 Two-Term Approximation

```
println(summarize(results[2]))
```

```
RitzResult:
  Domain:          [0.0, 1.0]
  Coefficients:    [0.19241192411920088, 0.1707317073170797]
  Minimized value: -0.024570912375790444
  Converged:       true
```

2.2.3 Three-Term Approximation

```
println(summarize(results[3]))
```

```
RitzResult:
  Domain:          [0.0, 1.0]
  Coefficients:    [0.18776681077849913, 0.1941430785544192, -0.02341137123737456]
  Minimized value: -0.024574009118017617
  Converged:       true
```

3 Comparison

```
Term count | Minimized J[y] | Coefficients
1          | -0.02314815    | [0.2777777777777778]
2          | -0.02457091    | [0.19241192411920088, 0.1707317073170797]
3          | -0.02457401    | [0.18776681077849913, 0.1941430785544192,
-0.02341137123737456]
```

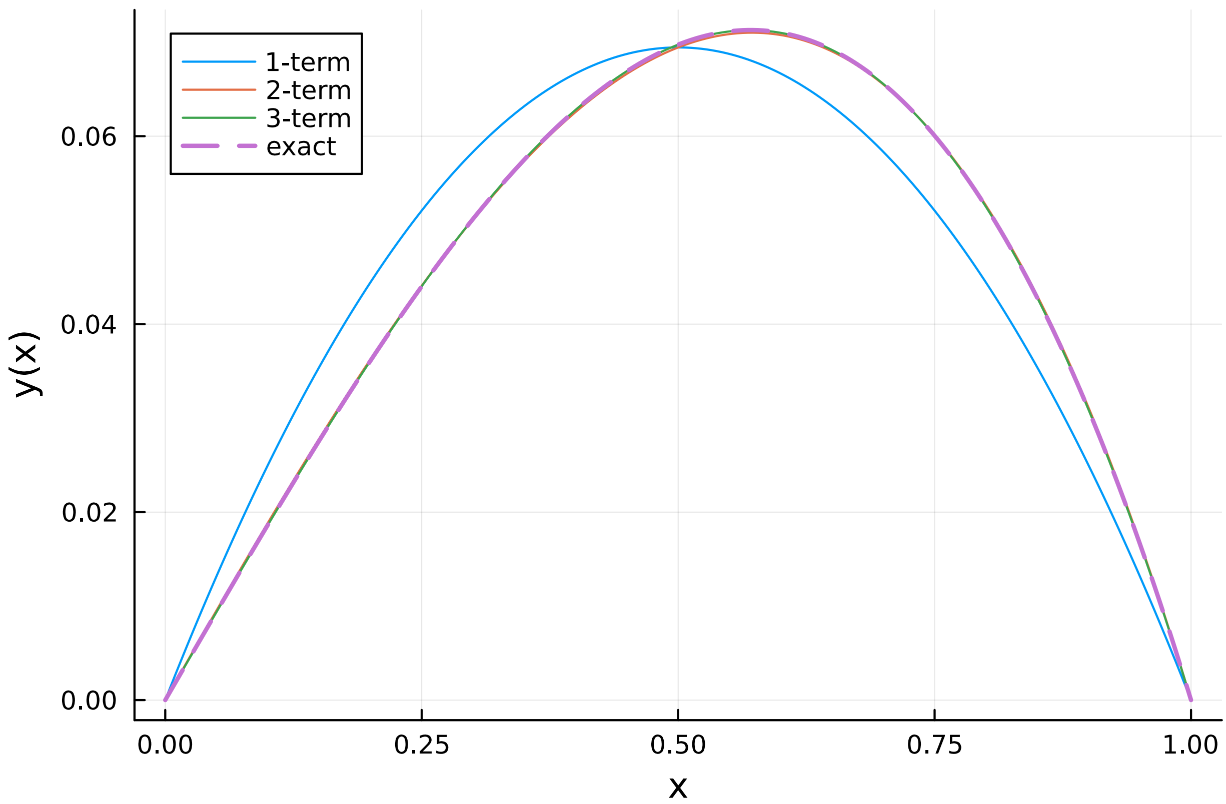


Figure 3: Ritz Approximations

4 Approximation Error

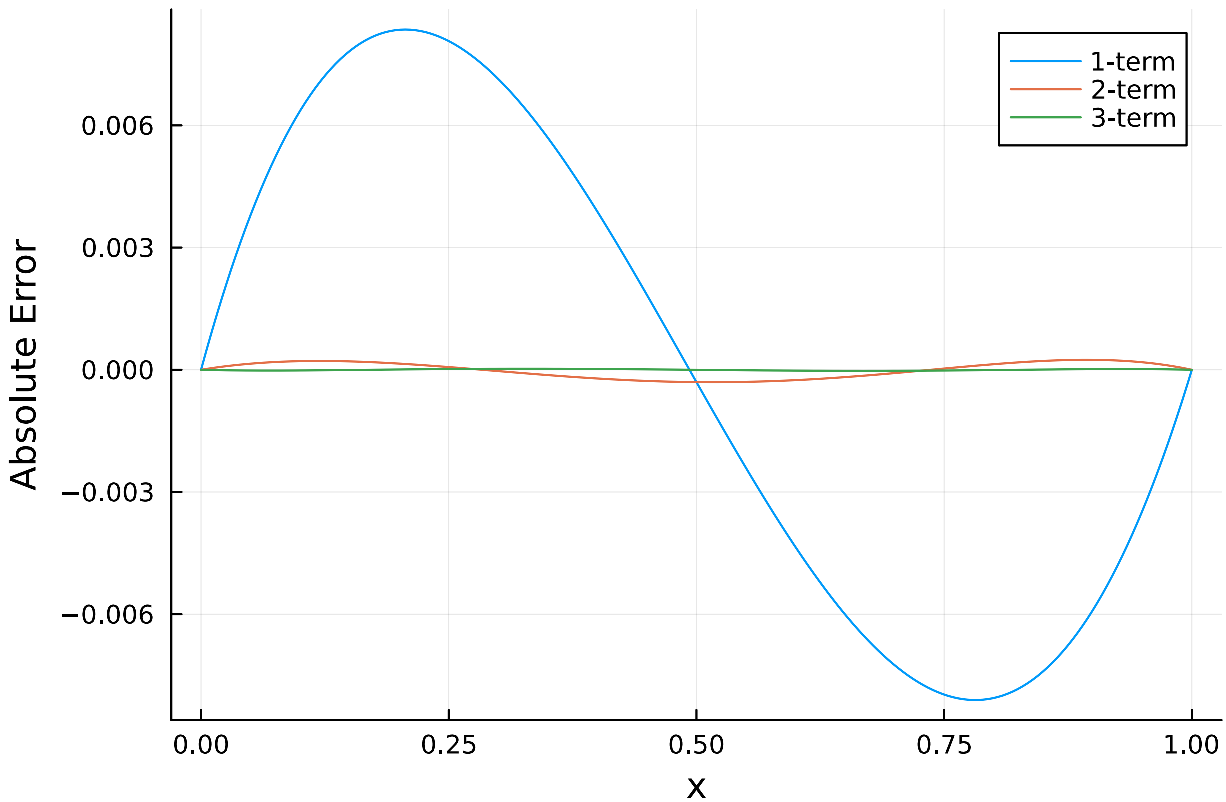


Figure 4: Pointwise error between Ritz approximations and the exact solution.